

Feeding a Changing World: Root Architecture and Rhizosphere Microbiome Insights in the Underutilised Cereal Crop *Eragrostis tef*

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1. Introduction

Global food systems depend heavily on just three crops — rice, wheat, and maize, while food demand is projected to rise by 62% by 2050. Under growing climate, geopolitical, and ecological pressures, diversification into underutilised crops is increasingly urgent.

Eragrostis tef (teff) is a promising candidate: drought- and waterlogging-tolerant, nutritionally rich (high protein, fibre, low GI, gluten-free), and a staple for ~89 million Ethiopians. Yet it remains understudied, with yields particularly constrained by lodging (bending at the stem or root), which accounts for 30–50% of yield losses.

Root system architecture (RSA) and rhizosphere microbiome composition are key determinants of lodging resistance and stress tolerance, yet both remain largely uncharacterised in teff. This study characterises RSA and rhizosphere microbiomes across multiple teff genotypes and root zones to identify strategies for improving lodging resistance and yield.

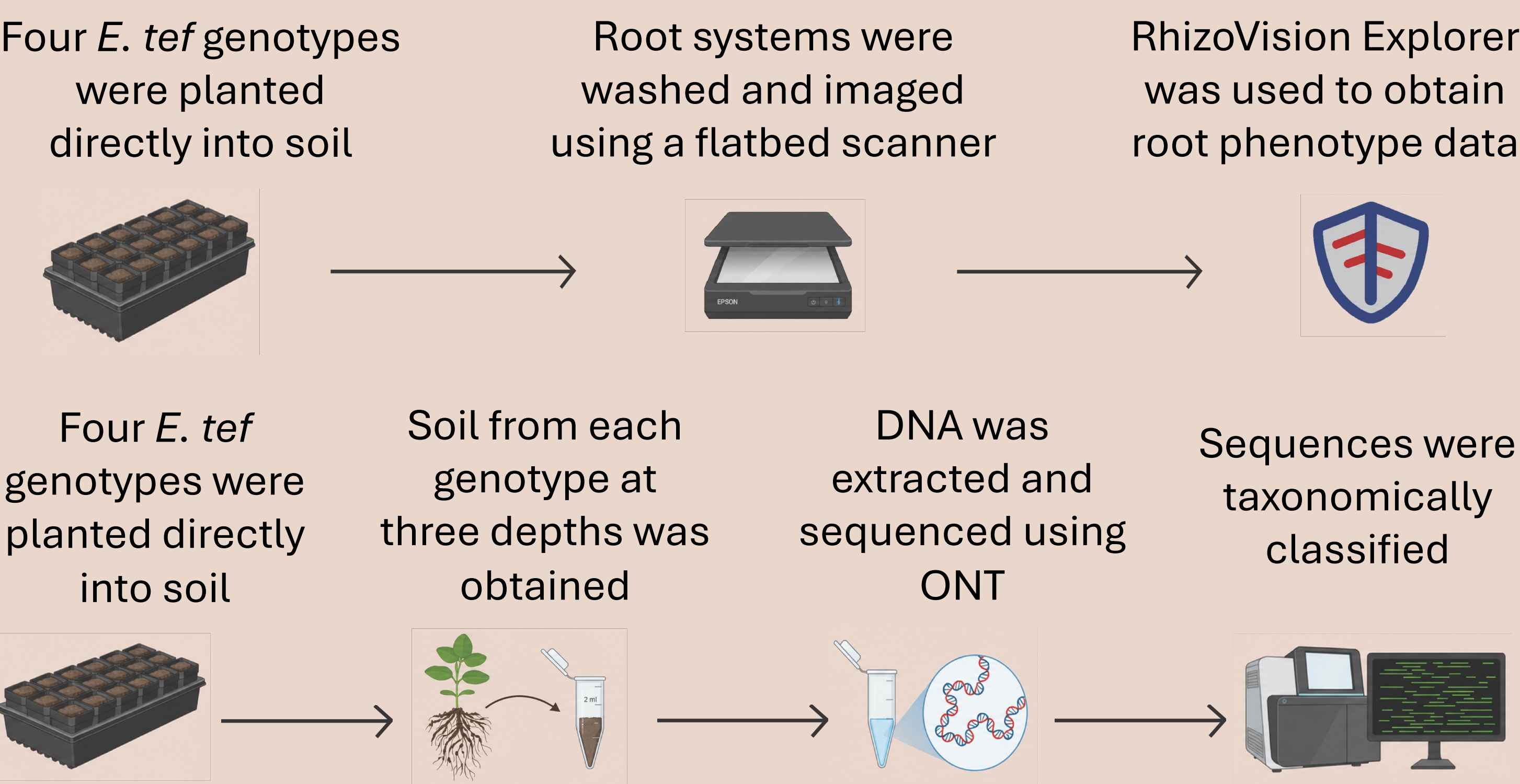
2. Aims

1. Is Oxford Nanopore (ONT) shotgun metagenomics a robust approach to exploring soil microbial communities?
2. Does the bulk soil differ from rhizosphere microbial communities?
3. Are there distinct microbial signatures between teff genotypes?
4. Are there distinct microbial signatures at different vertical root depths?
5. Are there genotype-specific differences in root system architecture?



Figure 1. Images of *Eragrostis tef* (A) plants in field (Günter Baron; African plants - A Photo Guide) and (B) seeds (Jan Bakker) and (C) diversity of *E. tef* genotypes (Gebru et al. 2020)

3. Methods



4. ONT correctly identifies taxa in a mock community

Pipeline was validated against the ZymoBIOMICS log-distribution mock community. The expected composition was recovered at genus level (Figure 2). The two fungal members (*Saccharomyces*, *Cryptococcus*) are absent because reads were classified against GTDB, a prokaryote-only database.

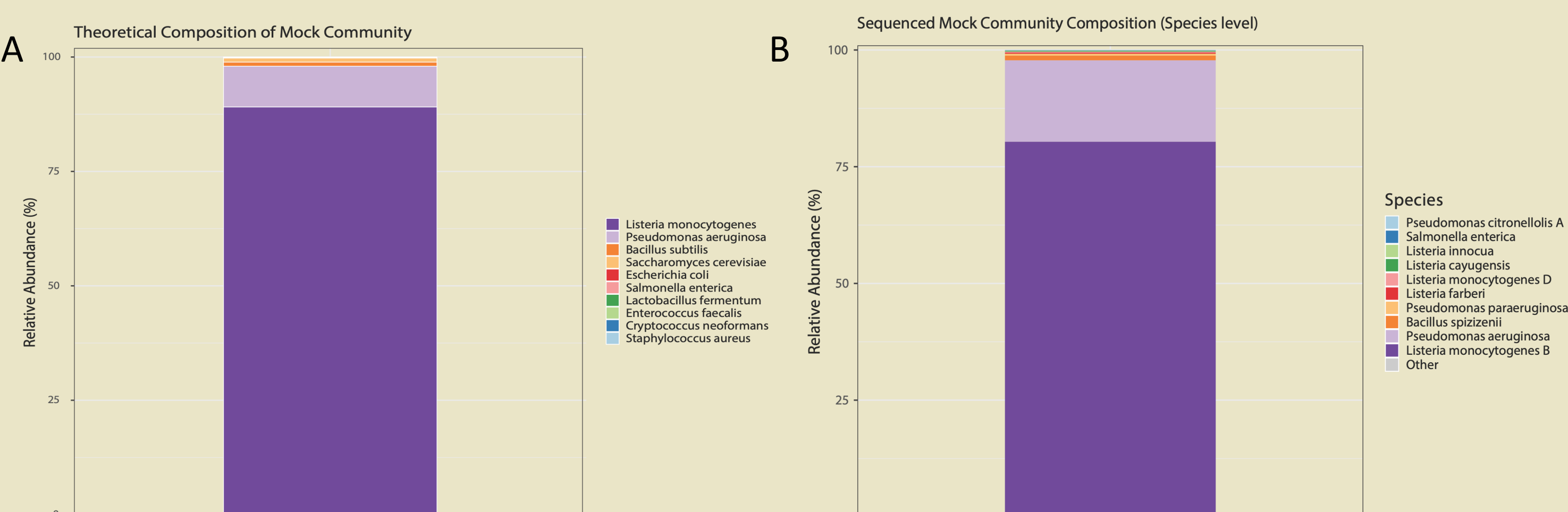


Figure 2. ZymoBIOMICS mock community (D6311) validates the pipeline. Expected (A) versus observed (B) composition at species level. Genus-level composition was accurately recovered; fungal members are absent as GTDB is prokaryote-only, and minor species-level splitting reflects read error and closely related references.

5. Bulk and rhizosphere soil showed distinct communities

Bulk soil appeared to host a markedly different community from the rhizosphere (Figure 3). Bulk soil was strongly dominated by *Streptomyces* whereas rhizosphere zones across all four genotypes supported a more even mix of multiple genera.

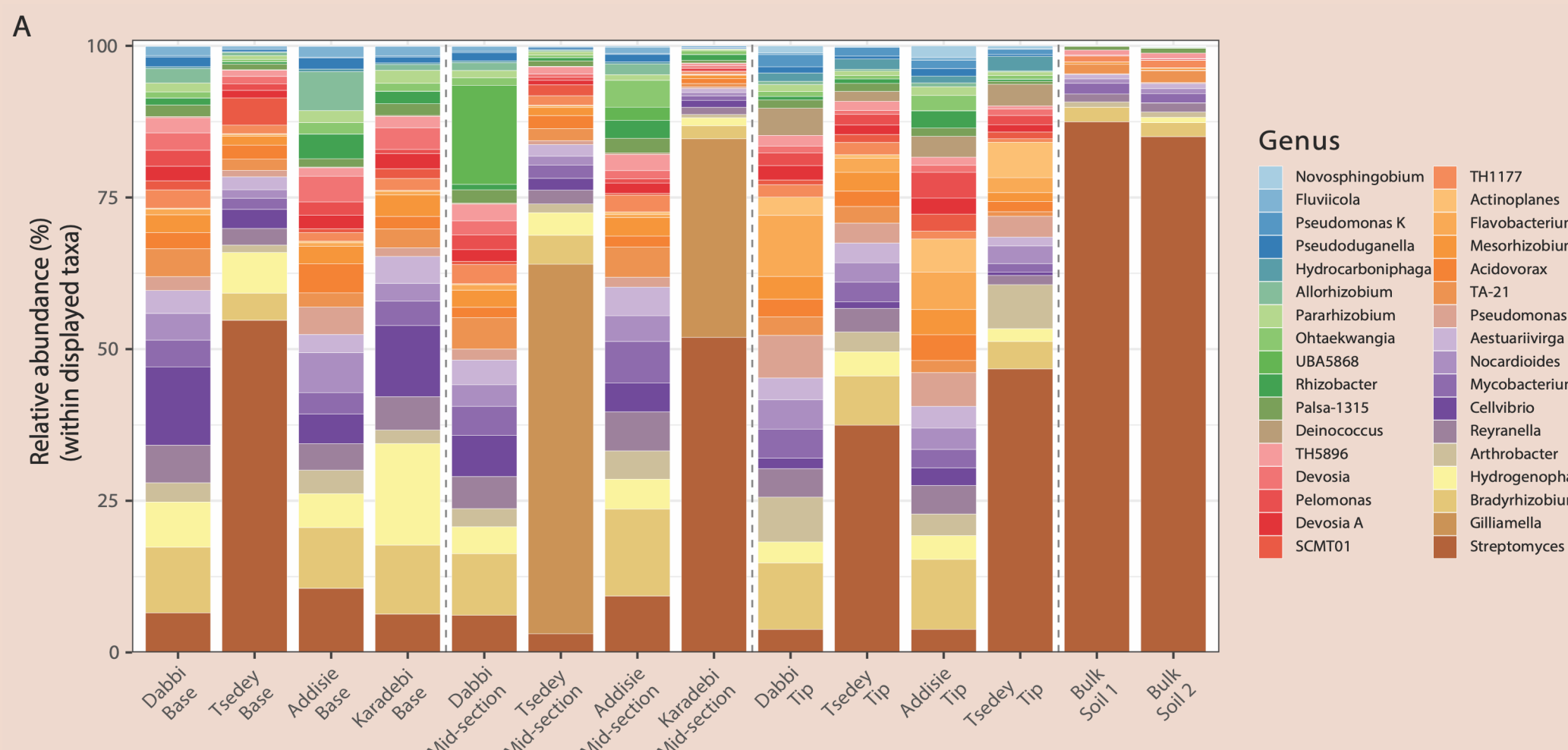


Figure 3. Genus-level composition across root zones and bulk soil. Relative abundance of top 1% of genera within each genotype × root-depth zone (replicates pooled) and two bulk-soil samples.

6. Distinct prokaryotic signatures between root depths, with weaker genotype effects

Principal coordinates analysis of prokaryotic communities revealed clear structuring by root depth (Figure 4). Genotypes were less clearly separated, suggesting a weaker genotype effect, and bulk-soil samples sat apart from the rhizosphere.

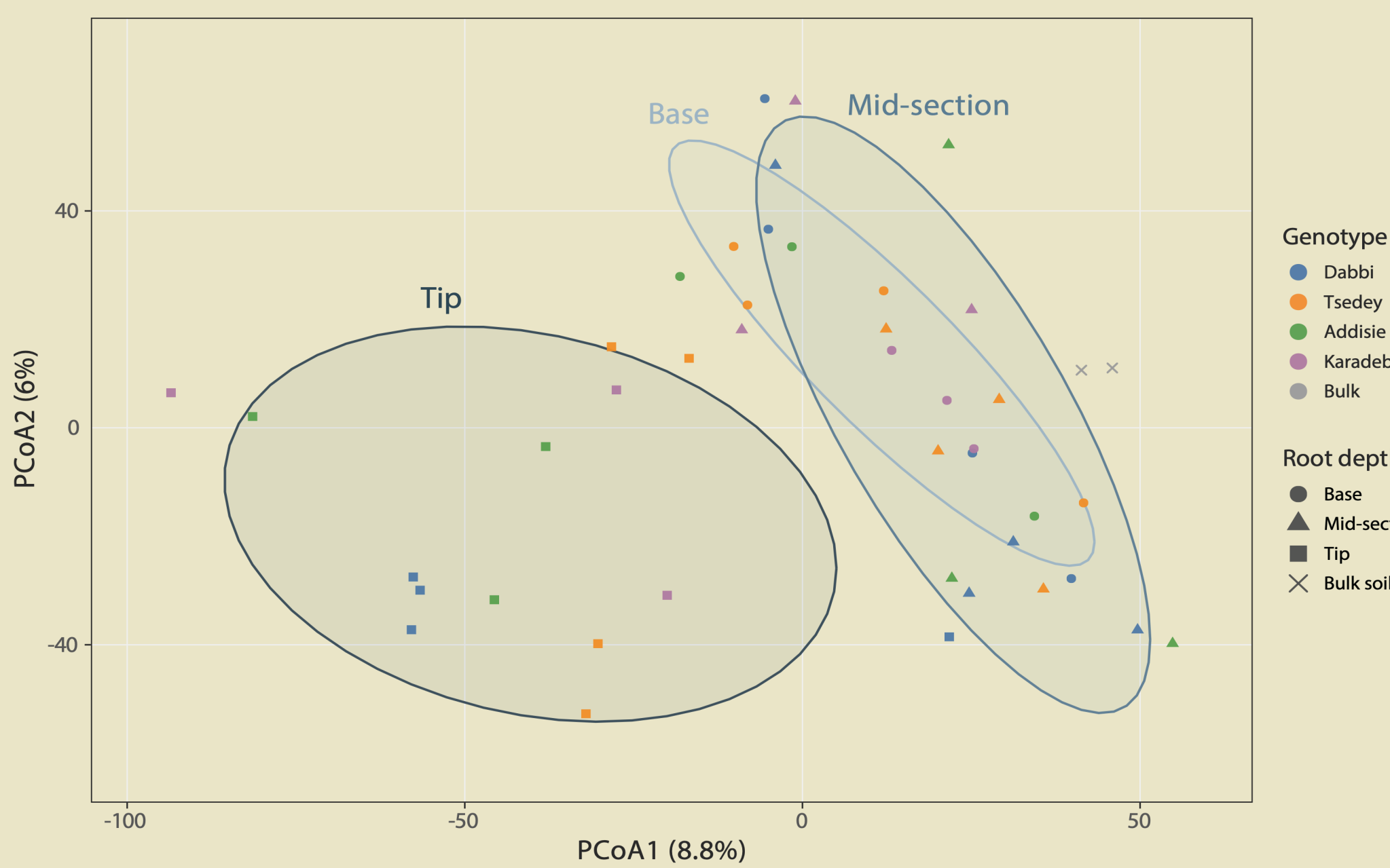


Figure 4. Principal coordinates analysis (PCoA) of microbial community composition based on Aitchison distances. Each point is one sample, coloured by genotype and shaped by root depth; bulk soil (X) is excluded from the ellipses. Shaded ellipses show the 68% level (≈ 1 SD) per root depth as a visual guide, not a statistical test. Axes give the percentage of total variation captured by each coordinate. Tip communities separate from base and mid-section, while genotypes are less distinct.

7. Two RSA phenotypes were identified

Root system architecture differed significantly among genotypes. Dabbi showed the widest, most spread-out system, while Tsedey had a narrower, steeper architecture (Figure 5). These represent potentially contrasting strategies: a steep, deep-foraging system (Tsedey) versus a wide, shallow-spreading one (Dabbi), both of which are plausibly advantageous for drought tolerance and lodging resistance.

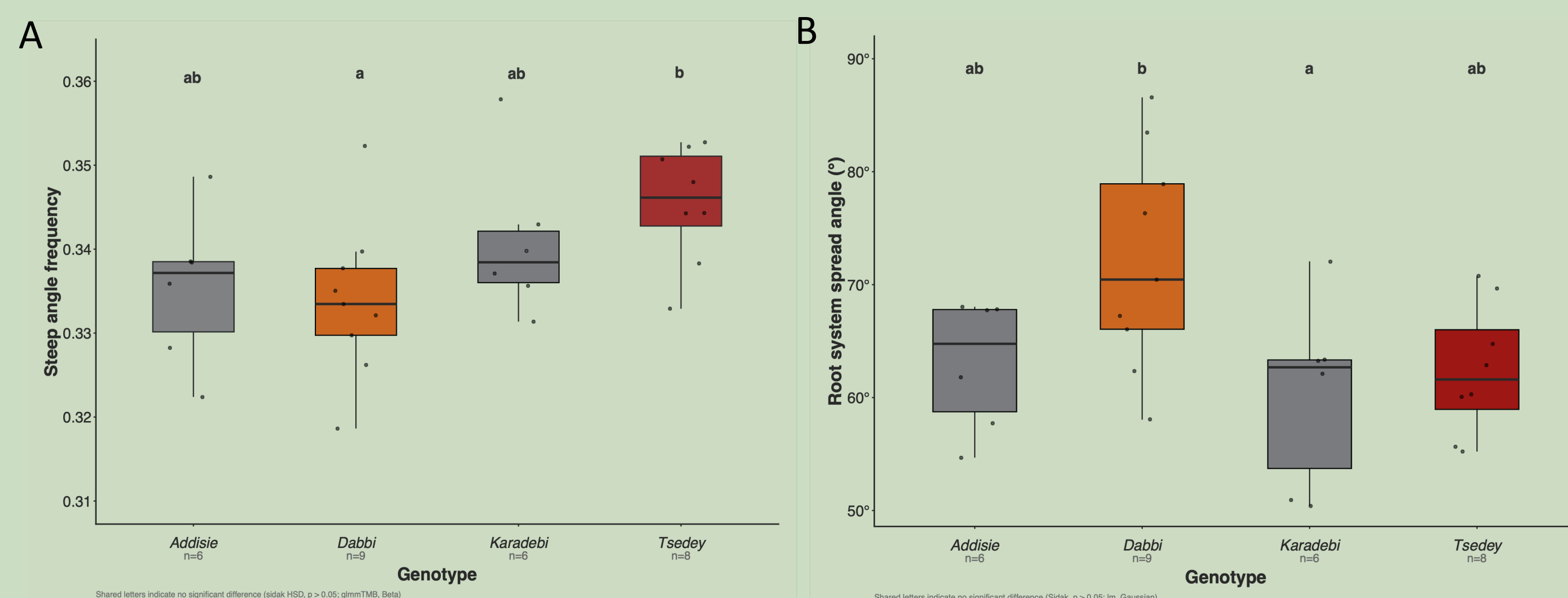


Figure 5. Root system architecture traits across four *Eragrostis tef* genotypes (six weeks post-germination). Boxplots show the distribution of (A) steep angle frequency and (B) root system spread angle (°). Points represent individual plant measurements. Compact letter display indicates pairwise significance (Sidak-adjusted, $p < 0.05$). Sample sizes per genotype are shown below each x-axis label.

8. Future directions

Future work should (i) validate these genotype- and depth-associated signatures under field conditions, (ii) test whether the divergent Tsedey/Dabbi root architectures confer measurable lodging or drought advantages, and (iii) evaluate targeted synthetic microbial communities for enhancing teff resilience and yield.